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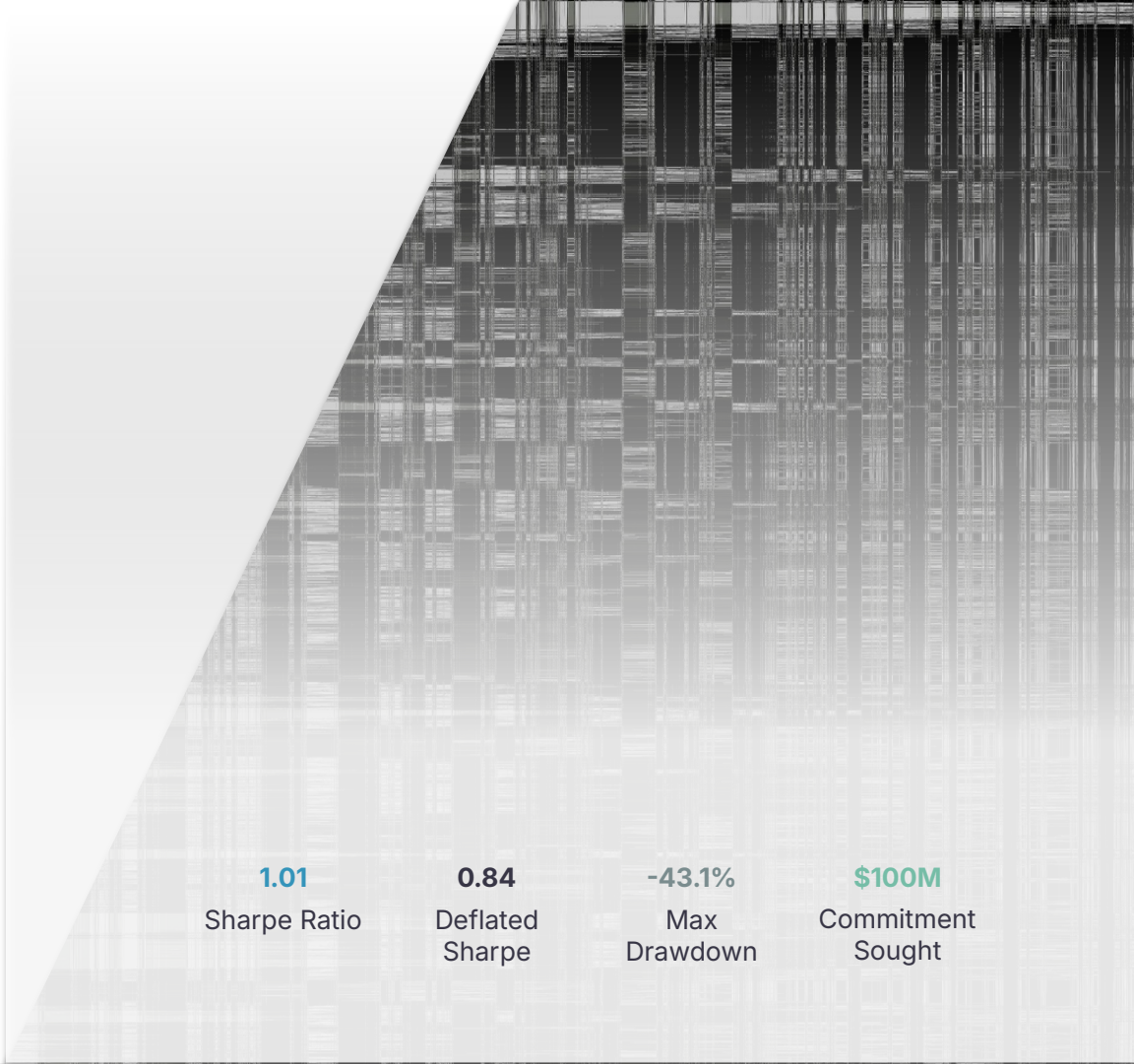
SUBSTRATE CAPITAL

Systematic long-short equity in the AI infrastructure complex

Asset Management – MSc Finance

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The Largest CapEx Cycle in Modern Market History

The CapEx flows into a definable set of public companies: chips, platforms, and the power grid that runs them.

$162B → $448B
Combined hyperscaler CapEx, 2022 to 2025

≈ $725B
Planned for 2026 alone, up 77% year on year

$5.3 trillion
Goldman Sachs estimate for the combined 2025-2030 period

Our Thesis: Why Now?

We do not take a view on whether AI succeeds. We systematically identify which infrastructure names the market has mispriced relative to the capital flowing through the AI-build-out.

The capex cycle is accelerating faster than analyst coverage and price discovery can keep pace, creating persistent cross-sectional mis-pricings that a systematic signal can harvest now.

"Investors stay calm as AI capex boom eclipses dotcom mania"
Reuters, May 28th, 2026

Sources: Epoch AI / SEC filings via Visual Capitalist (2022 & 2025 capex); Financial Times Q1-2026 earnings compilation ($725B, +77%); Goldman Sachs Global Investment Research ($5.3T, 2025–2030).

Motivation and relevance of the topic

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Cap-Weighted AI ETF

1. Long-only: profits only when sector rises, not specific sector leaders
2. Cap-weighted: effectively a mega-cap index proxy
3. Always fully exposed: no brakes in a crash

Substrate Capital

1. **Long-short: harvests the spread between the strongest and the weakest names**
2. **Conviction-weighted: by signal, not size, capped at 12% per holding**
3. **Regime overlay: cuts gross exposure to 50% in stress**

We take the sector beta deliberately and charge only for the long-short selection on top: positive, but over this sample not yet statistically significant (alpha t = 0.67). The market-neutral variant isolates it for fee-sensitive investors.

Three Connected Layers in CapEx Flows

CAPEX SPENDERS

Hyperscalers:

MSFT · GOOGL · META · AMZN · ORCL · CRM · NOW · SNOW · PLTR · DDOG · MDB · NET · CRWD · ZS · OKTA · AI · U · PATH · S · GTLB

≈ $725B 2026E capex

RECEIVERS: COMPUTE

Semiconductors: build the chips

NVDA · AMD · AVGO · MRVL · INTC · AMAT · LRCX · KLAC · ASML · TSM · ARM · MU · MPWR · ON · QCOM · MCHP · NXPI · ADI · TXN · SMCI

RECEIVERS: COMPUTE

Semiconductors: supply the electricity

VRT · ETN · EMR · JCI · CARR · CEG · VST · NRG · TLN · AES · NEE · DUK · SO · MOD · FLR · J · EXP · PWR

58 Names:
One dollar of AI capex touches every link: hyperscalers spend it, chipmakers turn it into compute, power names supply the electricity that compute consumes. That shared structural driver, not a generic AI-stock screen, is why they form one universe. The model captures this today through common momentum and macro; encoding the explicit supplier-customer links is the next build.

Fifteen Features; Price-Momentum Leads

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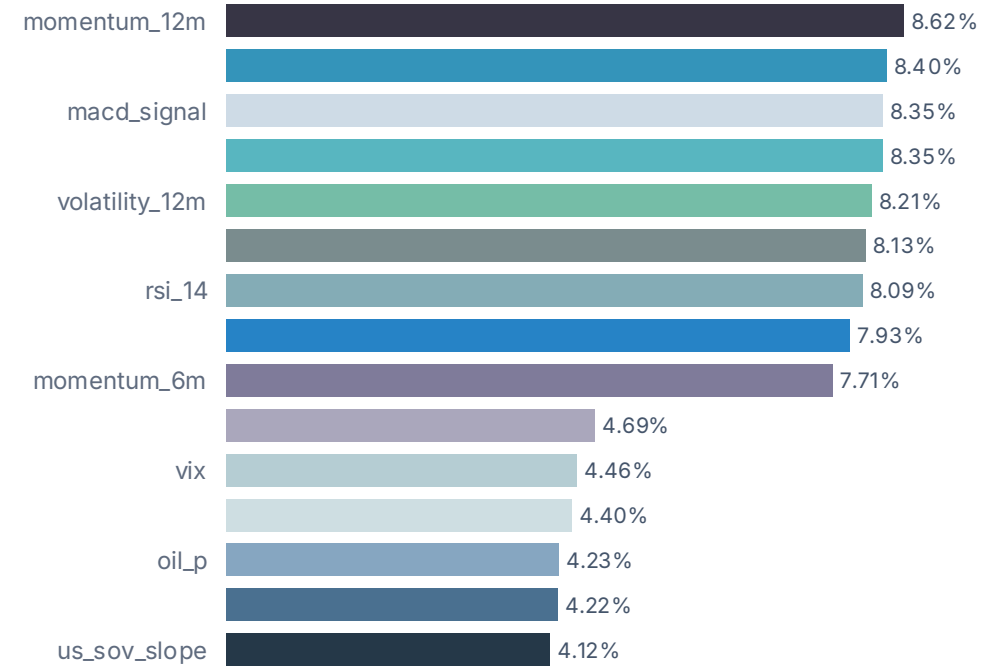
Technical | 9

- Momentum 1m, 3m, 6m, 12m
- 12-Month Volatility
- RSI (14)
- MACD Signal
- Bollinger %B
- Volume Z-Score

Macro | 6

- VIX
- 10-Year & 5-Year Treasury Yields
- Term Spread (10Y– 5Y)
- WTI Crude
- S&P 500 Return

Random Forest Features Importance

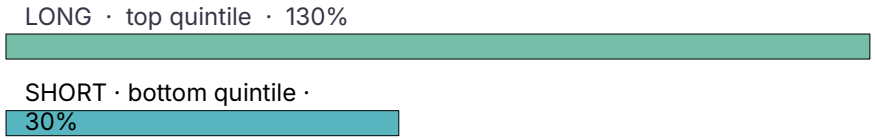


Target

Outperform the cross-sectional median next month (binary). Fundamentals excluded to prevent forward bias.

A 130 / 30 systematic long-short

Exposure



1. Rank all 58 names monthly by predicted probability
2. Probabilities smoothed over 3 months to cut turnover
3. Equal weight within each leg, 12% hard single-name cap
4. Monthly rebalance — matched to the signal's cadence

Exposure

Asymmetry
The model ranks winners more reliably than losers, so we lean long: 130% long, 30% short.

Conviction
A large long book where the signal is strong, a small short overlay where it's weaker.

Evidence
The top-vs-bottom decile gap holds across the out-of-sample test, not just in-sample.

The 30% short funds the long extension and trims net market exposure.

Control Codes

Regime Overlay

2-state HMM on VIX + term spread cuts gross exposure to 50% in stress.

Hard Position Cap

12% maximum per name, enforced in construction; cluster exposure monitored.

Tail Risk

Monthly VaR(95%) −10.4% · CVaR(95%) −14.5% · max drawdown −43.1%.

Stress Test

Drawdown

−19.48% — Peak-to-trough drawdown during the COVID crash of February to April 2020, the strategy's sharpest stress event in the test period.

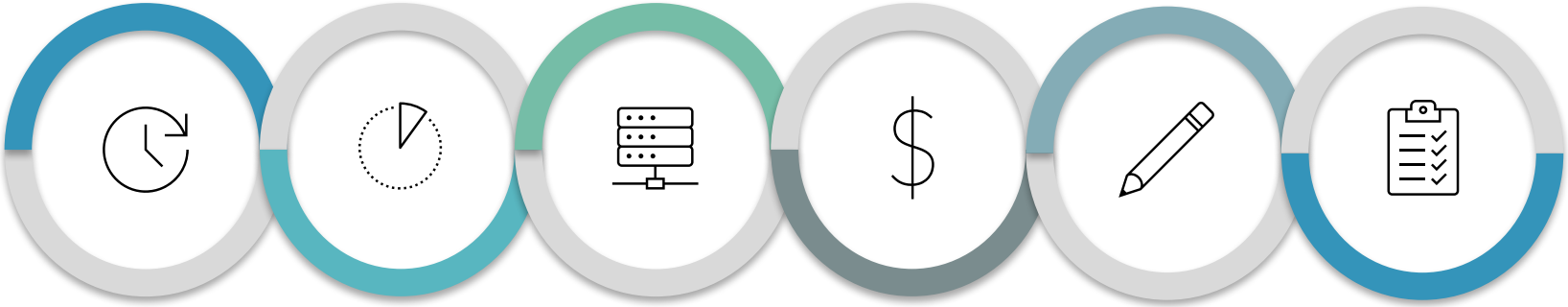
De-risking

50% — The regime overlay read the volatility spike and cut gross exposure in half, taking risk down before the worst of the drawdown hit.

Recovery

2 months — The book recovered to its prior high within two months, far faster than the broad market retraced the same crash.

Backtesting Methodology



1. Walk-Forward, Expanding Window

36-month minimum training, refit yearly, predict forward only

2. Two-Month Embargo

A gap between train and test windows so nothing leaks (López de Prado)

3. Out-of-sample: 2014–2026

147 months, ~6,900 monthly predictions

4. Costs Charged Per Side

15 bps per side on turnover, about 0.15% a month

5. Deflated Sharpe Ratio

Corrects Sharpe for how many variants we tried (LdP 2014)

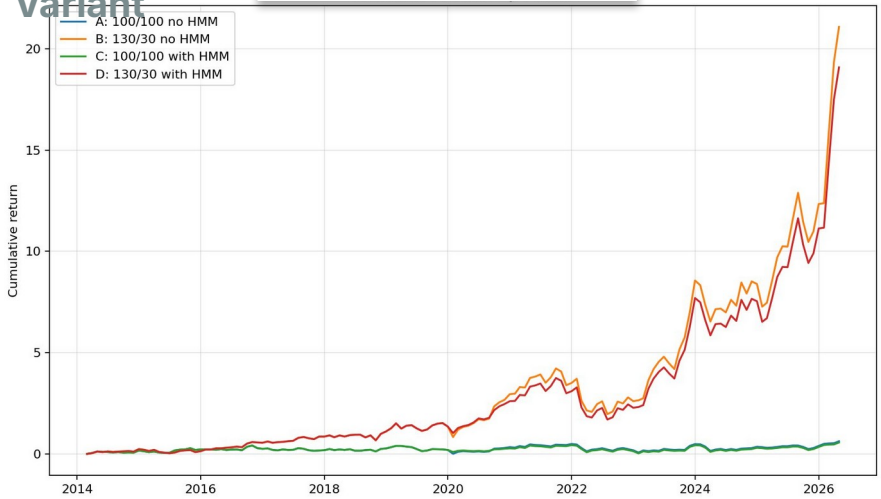
6. Every Variant Shown

No cherry-picking; survivorship re-run included

Strategy

Variant	Sharpe	Deflation	Max DD
Neutral, no overlay	0,22	0,09	-30%
130/30, no overlay	1,05	0,87	-43%
Neutral + Overlay	0,22	0,08	-28%
130/30 + Overlay	1,01	0,84	-43%

Cumulative returns by Variant

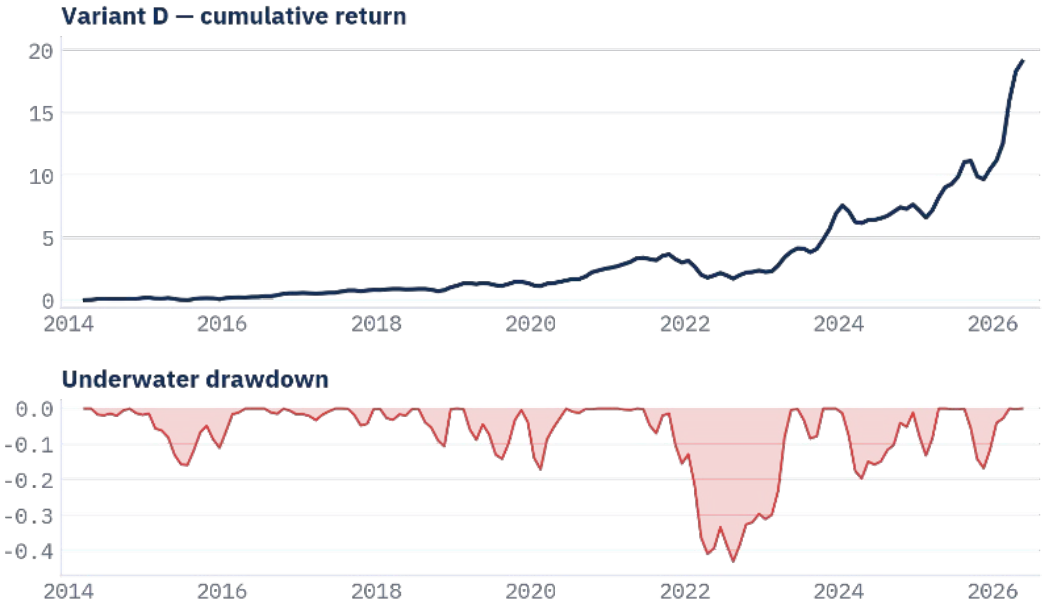


We tested four structural variants and report all of them. Market-neutral isolates the pure selection signal; 130/30 adds the thematic beta we take deliberately.

Results

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We pitch D: the only book that survives deflation (DSR 0.84 and carries a regime brake that held the COVID drawdown to −19.48%. Cumulative simulated return of 1907.75% over 147 months. A deflated Sharpe of 0.84 after correcting for the variants tested, sits in the institutionally acceptable range.



Sharpe Ratio of 1.01

Strong risk-adjusted performance (>1.0 is generally considered attractive)

Deflated Sharpe 0.84

Performance remains positive after accounting for backtest overfitting risk.

CAGR 27.7%

High annual return, indicating strong historical growth

Hit Rate 66.67%

Most trades are profitable, suggesting reliable signal quality (above 50% indicates consistent trade accuracy)

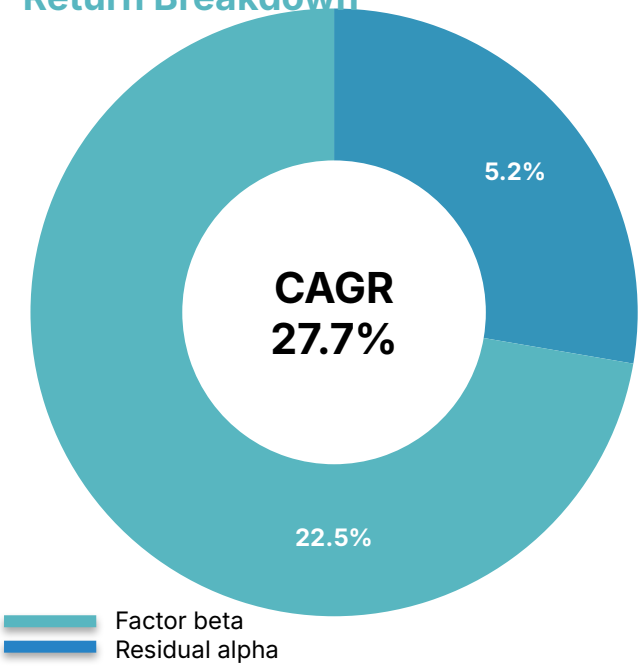
Max DD 43.1%

Large historical loss period, highlighting meaningful downside risk.

Turnover 44.2% (One-way)

Moderate trading intensity with manageable cost impact.

Return Breakdown



Multi-Factor regression

Annualized alpha	**5.2%**
Alpha t-stat	0.67
SOXX beta	0.73
S&P beta	0.12
R^2	0.566

We take the sector beta deliberately and charge only for the residual. Alpha is ~5% but not statistically significant over 147 months (t = 0.67); our stronger evidence of signal is the monotonic decile spread, which survives the survivorship re-run.

Signal Survival

The concern

Our 58 names are today's AI winners, so the backtest could just be reflecting hindsight.

The test

We re-ran the full strategy on only the 37 names investable before 2012, dropping every late IPO and spinoff.

The result

Sharpe barely moves (1.01 → 1.00), deflated Sharpe holds at 0.84, and the decile spread widens on the established names.

Survivorship Robustness

	Full 58	Pre-2012 (37)
Sharpe	1.01	1.00
Deflated Sharpe	0.84	0.84
CAGR	27.7%	24.1%
Max Drawdown	-43.1%	-22.5%
Decile 10 - Decile 1	+0.5%	+1.8%

Our Services and Terms

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Management Fee

1.5% annual



Performance Fee

15% over the 5% hurdle rate



Liquidity

Monthly, 60-day notice period



Lockup

None

Capacity Limit : $250M AUM

Large-cap universe, ADV > $200M · conservative participation at full size · assignment-level estimate, not a fitted impact model

Disclosures: Simulated, Repeatable

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All Results are Simulated Backtests

No live trading. No paper-trading results mixed in.

Variants Tested

Four structural variants shown; deflated Sharpe reported through N = 10.

Look-ahead Controls

Walk-forward, 2-month embargo, fundamentals excluded, HMM frozen on pre-OOS data.

Survivorship

Disclosed as the primary limitation; robustness re-run on the pre-2012 subset included.

AI Tools and Workflow

Built with specialized sub-agents with scoped system prompts, orchestrated in a deterministic workflow that loops back at evaluation checkpoints for review.

Data Sources

yfinance public API — prices, volume, macro, sector ETFs.

THANK YOU

Substrate Capital

Systematic long-short equity in the AI infrastructure complex



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